# FreshBasket AWS Cloud Migration

Cloud Architecture & Deployment Report


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|---|---|
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## Document Structure

## PART 1 — AWS SYSTEM ARCHITECTURE (Deliverable 1)

### 1.1 Architecture Overview

The proposed architecture transitions FreshBasket from a single desktop PC,  a critical single point of failure, to a robust, scalable cloud environment on Amazon Web Services. The design is multi-tier and highly available, centred on AWS Elastic Beanstalk for application orchestration and Amazon RDS for managed data persistence.

Incoming HTTP traffic enters through an Application Load Balancer (ALB), which distributes requests across a fleet of EC2 instances managed by an Auto Scaling Group (ASG). Instances are spread across two distinct Availability Zones within a custom Virtual Private Cloud (VPC), guaranteeing fault tolerance at the compute tier. The application communicates with an Amazon RDS MySQL database in Multi-AZ mode, providing synchronous replication and automatic failover. Amazon SNS delivers email alerts for environment health changes and scaling events.

### 1.2 Architecture Diagram

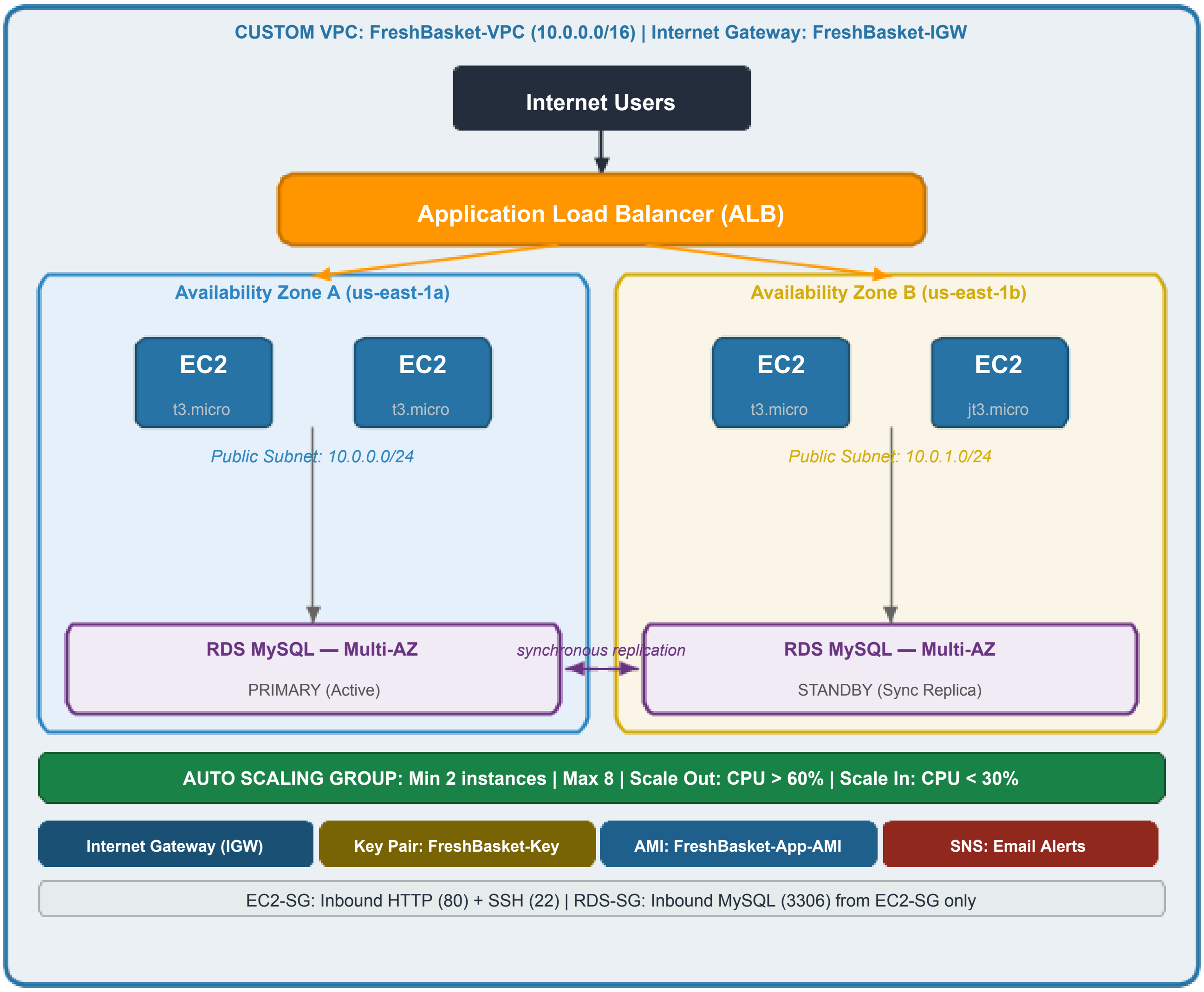



Figure 0: FreshBasket AWS Architecture — Internet Users → ALB → EC2 Auto Scaling Group (min 2, max 8 across 2 AZs) → RDS MySQL Multi-AZ (synchronous replication, automatic failover).

## 1.3 Justification of AWS Components

| Component | Scalability Justification | Disaster Recovery Justification |
|---|---|---|
| **Custom VPC (10.0.0.0/16)** | /16 CIDR provides 65,536 IPs — ample room for subnet growth as the business expands. | Multi-AZ subnets ensure network-level redundancy; an AZ outage does not affect the other subnet. |
| **Application Load Balancer** | Auto-scales its own capacity; distributes requests evenly across all healthy EC2 instances. | Continuous health checks on EC2 targets; routes traffic only to healthy instances across both AZs. |
| **Auto Scaling Group (2–8)** | Adds instances when CPU > 60%, removes when < 30% — directly addresses the TV-surge risk without over-provisioning. | Maintains minimum 2 instances across 2 AZs at all times; auto-replaces any failed instance. |
| **Custom AMI (FreshBasket-App-AMI)** | Pre-installed dependencies reduce instance boot time during scale-out, enabling faster response to traffic spikes. | Every ASG-launched instance is identically configured — eliminates drift and ensures consistent behaviour. |
| **Amazon RDS MySQL (Multi-AZ)** | Managed service handles storage auto-scaling and IOPS; read replicas can be added for future read-heavy loads. | Synchronous standby replica in a separate AZ; automatic failover in 60–120 s with zero data loss. |
| **EC2 Security Group** | Stateful firewall scales automatically with the ASG — no manual reconfiguration as instances are added. | Restricts inbound to ports 80 (HTTP) and 22 (SSH) only, minimising the attack surface. |
| **RDS Security Group** | Scales with the database tier without manual intervention. | MySQL (3306) access restricted exclusively to the EC2 SG — database never reachable from the internet. |
| **Amazon SNS (Notifications)** | Alerts operators to scaling events, enabling proactive capacity planning. | Immediate health-change notifications allow rapid human response to issues automated recovery cannot resolve. |

## 1.4 Statement of Assumptions

| Category | Details |
|---|---|
| **AWS Region** | All resources are deployed in ap-southeast-2 (Sydney) to minimise latency for FreshBasket's Sydney-based user base. |
| **Instance Type** | EC2 instances use t3.micro — CPU-burstable, suitable for variable web traffic. |
| **Database Instance** | RDS uses db.t3.small with 20 GB allocated storage and MySQL 8.0. |
| **Application Architecture** | The web application is stateless. Session data is stored client-side (cookies) or in RDS, so any EC2 instance can serve any request. |
| **Security Strategy** | The database requires no public internet access. All database interactions occur through the EC2 application tier. SSH (port 22) is open from any IP for development. |
| **IAM Constraints** | The AWS Academy Learner Lab provides two pre-built identities: LabRole (Beanstalk service role) and LabInstanceProfile (EC2 instance profile). No custom IAM roles can be created. |
| **Key Pair** | A single RSA key pair (FreshBasket-Key) is shared across all EC2 instances for SSH access. |
| **DB Credentials** | RDS connection details are injected via Elastic Beanstalk environment properties: RDS_HOSTNAME, RDS_PORT, RDS_DB_NAME, RDS_USERNAME, RDS_PASSWORD. |

## 1.5 Complete List of AWS Services Used

| # | AWS Service | Purpose in Architecture |
|---|---|---|
| 1 | **Amazon VPC** | Custom isolated network (10.0.0.0/16) with two public subnets |
| 2 | **Internet Gateway** | Provides internet connectivity to VPC resources |
| 3 | **Route Table** | Routes 0.0.0.0/0 traffic to the Internet Gateway for both public subnets |
| 4 | **AWS Elastic Beanstalk** | PaaS orchestration of the full application environment (EC2, ALB, ASG) |
| 5 | **Amazon EC2** | Compute instances running the PHP web application |
| 6 | **Custom AMI** | Pre-configured image (FreshBasket-App-AMI) for consistent, fast instance launches |
| 7 | **EC2 Key Pair** | RSA key pair (FreshBasket-Key) for SSH authentication to all instances |
| 8 | **Application Load Balancer** | HTTP traffic distribution across EC2 instances in both Availability Zones |
| 9 | **Auto Scaling Group** | Dynamic capacity management — 2 to 8 instances, CPU-based scaling triggers |
| 10 | **Amazon RDS (MySQL)** | Managed relational database with Multi-AZ deployment for high availability |
| 11 | **Security Groups** | Stateful firewalls for EC2 (HTTP/SSH) and RDS (MySQL from EC2-SG only) |
| 12 | **Amazon SNS** | Email notifications for environment health changes and scaling events |

# PART 2 — AWS IMPLEMENTATION EVIDENCE (Deliverable 2)

Figure 1. Custom VPC

VPC overview showing VPC name FreshBasket-VPC, CIDR 10.0.0.0/16, two subnets in different AZs, and route table with 0.0.0.0/0 routed to the Internet Gateway.

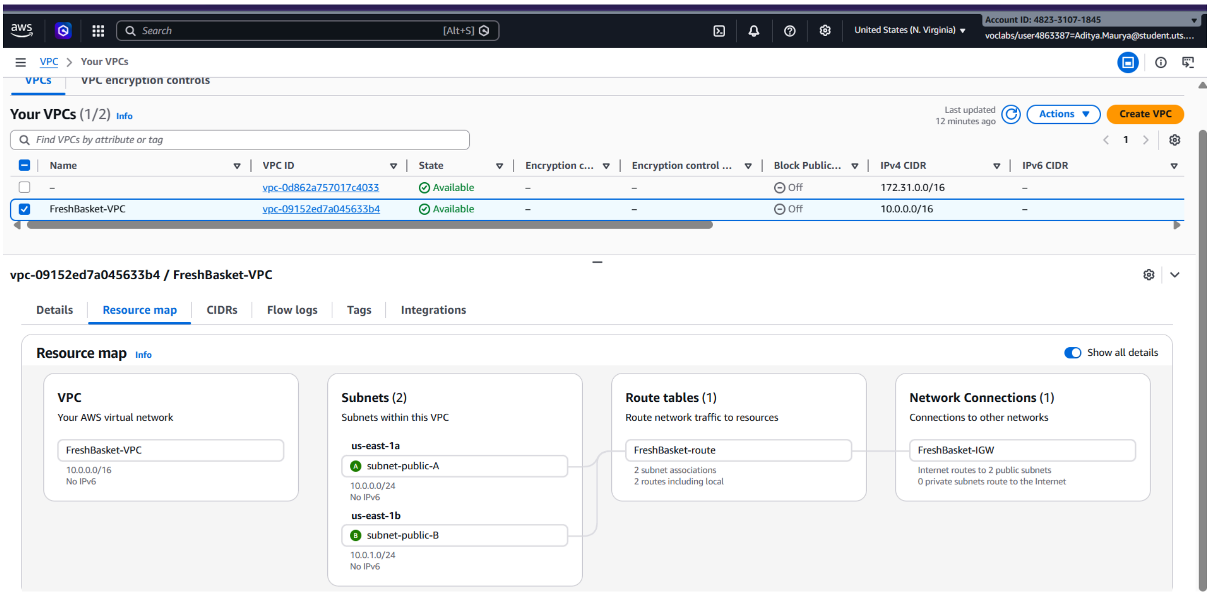



Figure 2. Route Table

Route table view confirming that both subnets are public (with a default route to an Internet Gateway)

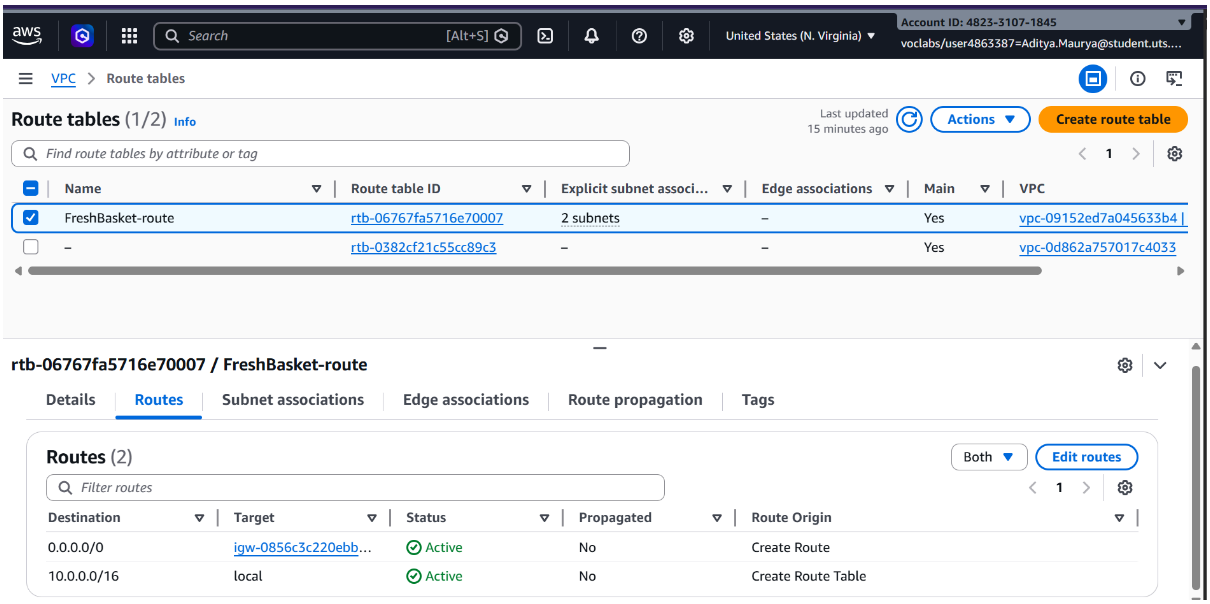


Figure 3. EC2 Security Group

Inbound rules of the FreshBasket-SG showing port 22 (SSH) and port 80 (HTTP) open from 0.0.0.0/0.

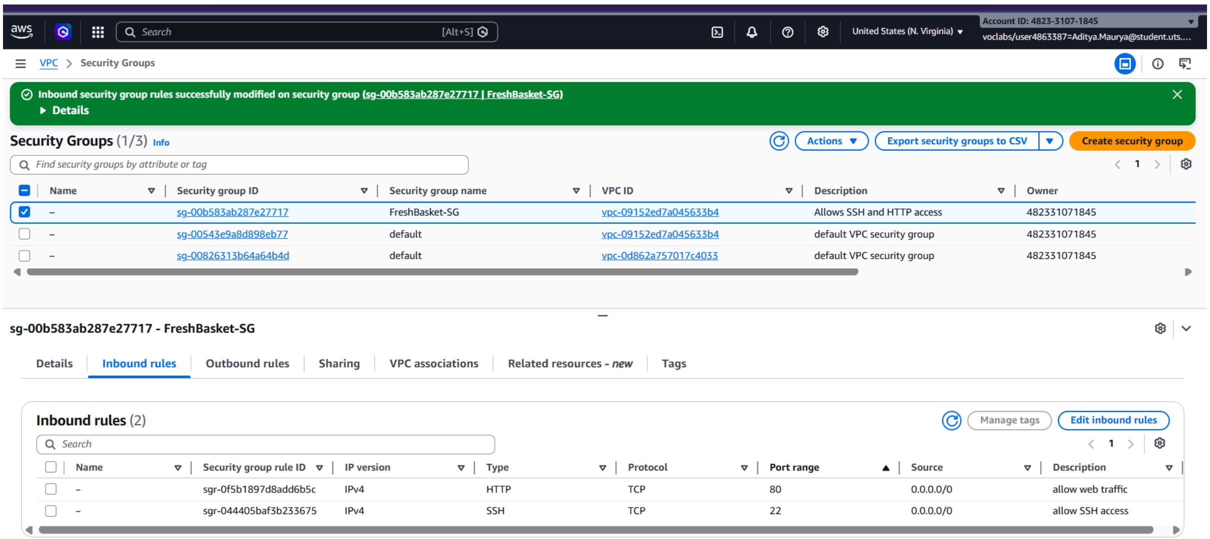



Figure 4. RDS Group

Inbound rules of FreshBasket-DB-SG showing port 3306 (MySQL) with source set to the EC2 security group ID — not a CIDR range.

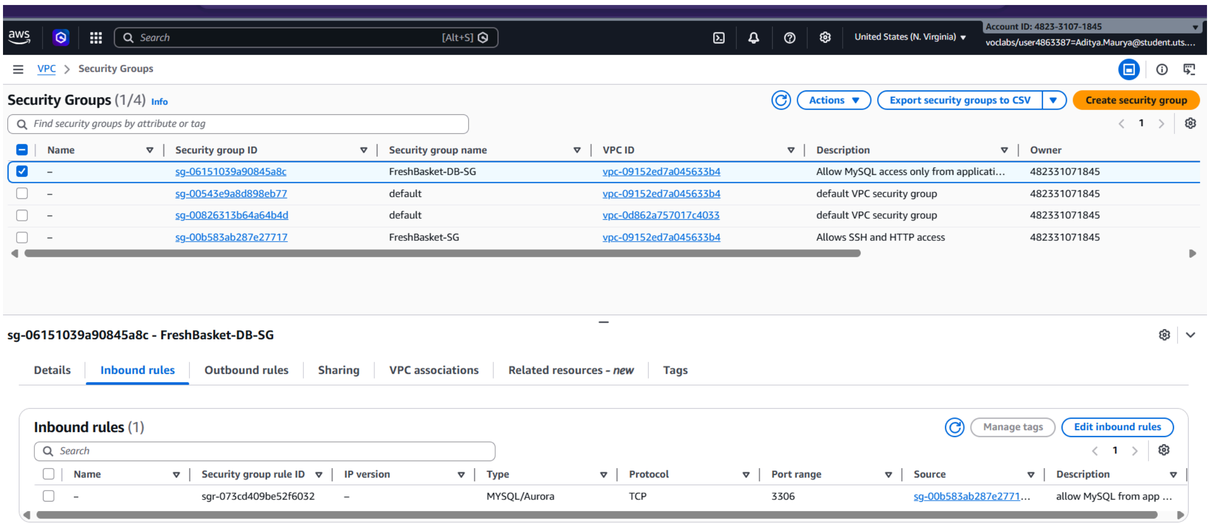


Figure 5. Custom EC2 Key Pair

EC2 → Key Pairs page showing the custom key pair FreshBasket-Key created for this deployment.

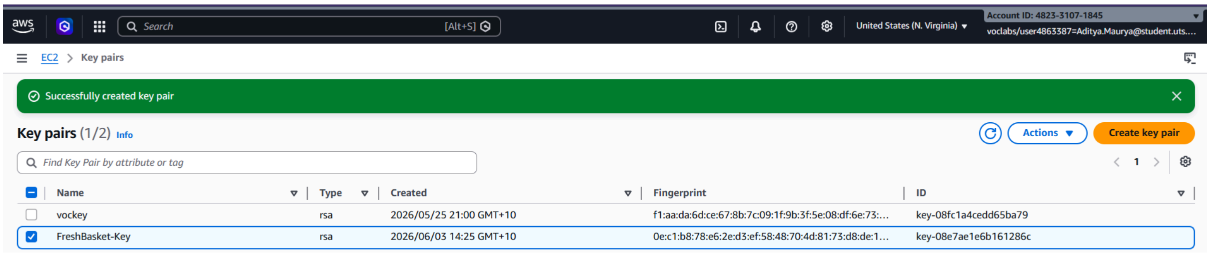



Figure 6. Custom AMI

EC2 → AMIs page (filter: Owned by me) showing FreshBasket-App-AMI with status Available.

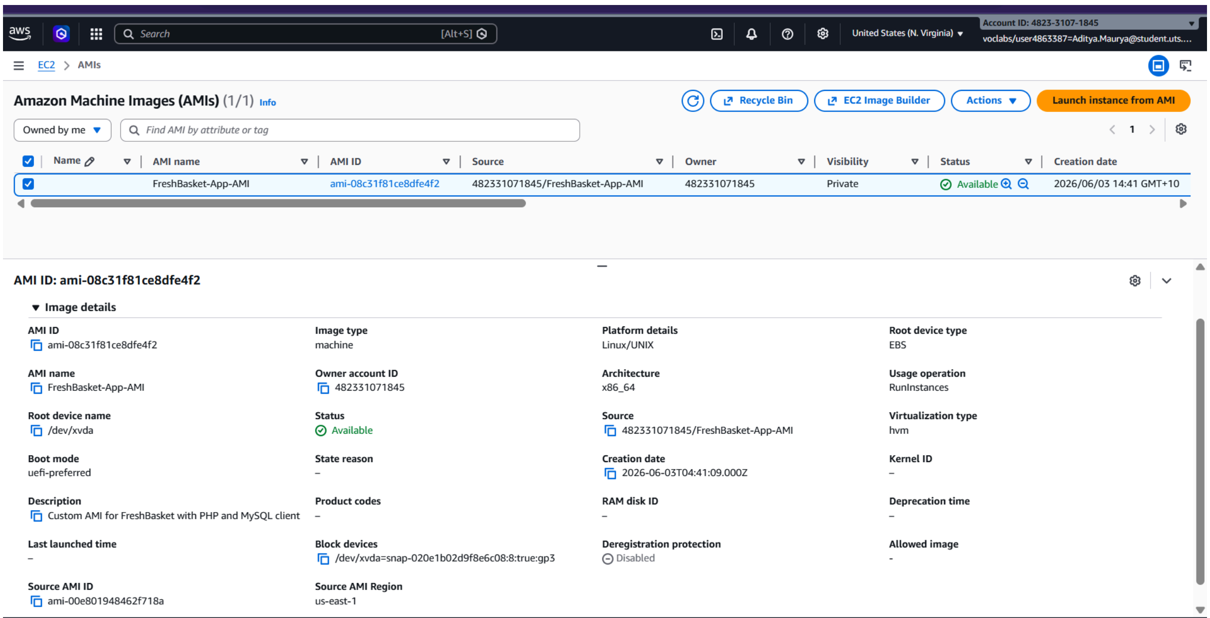


Figure 7. RDS Multi AZ Configuration

RDS instance summary page with Multi-AZ field clearly showing Yes and the two Availability Zones in use.

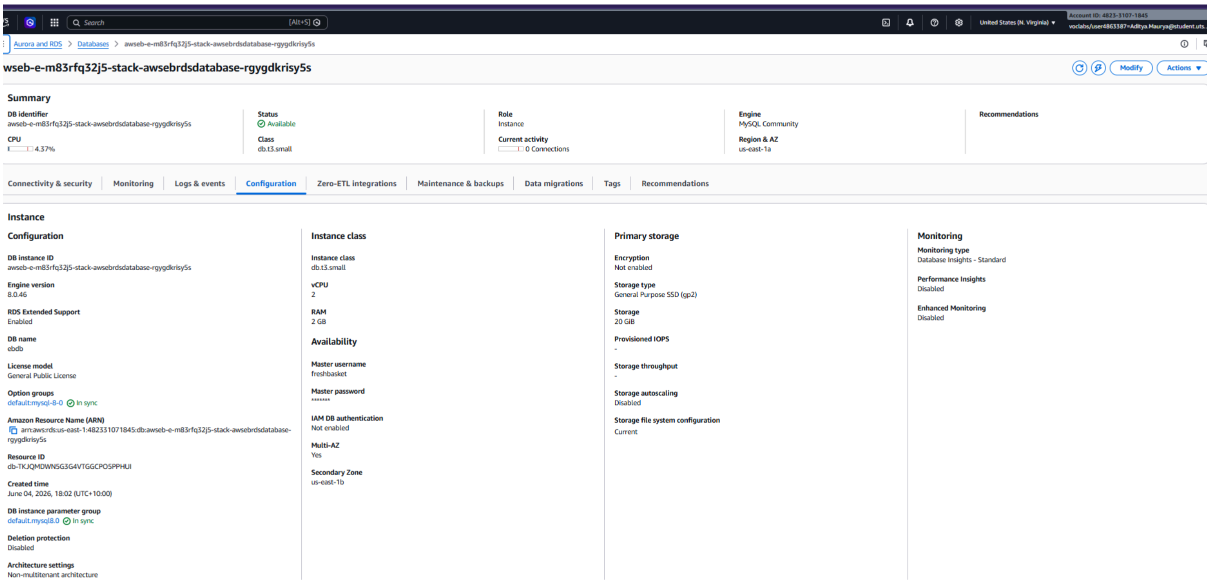


Figure 8. Elastic Beanstalk Dashboard

Beanstalk environment overview showing health status Ok and the environment URL.

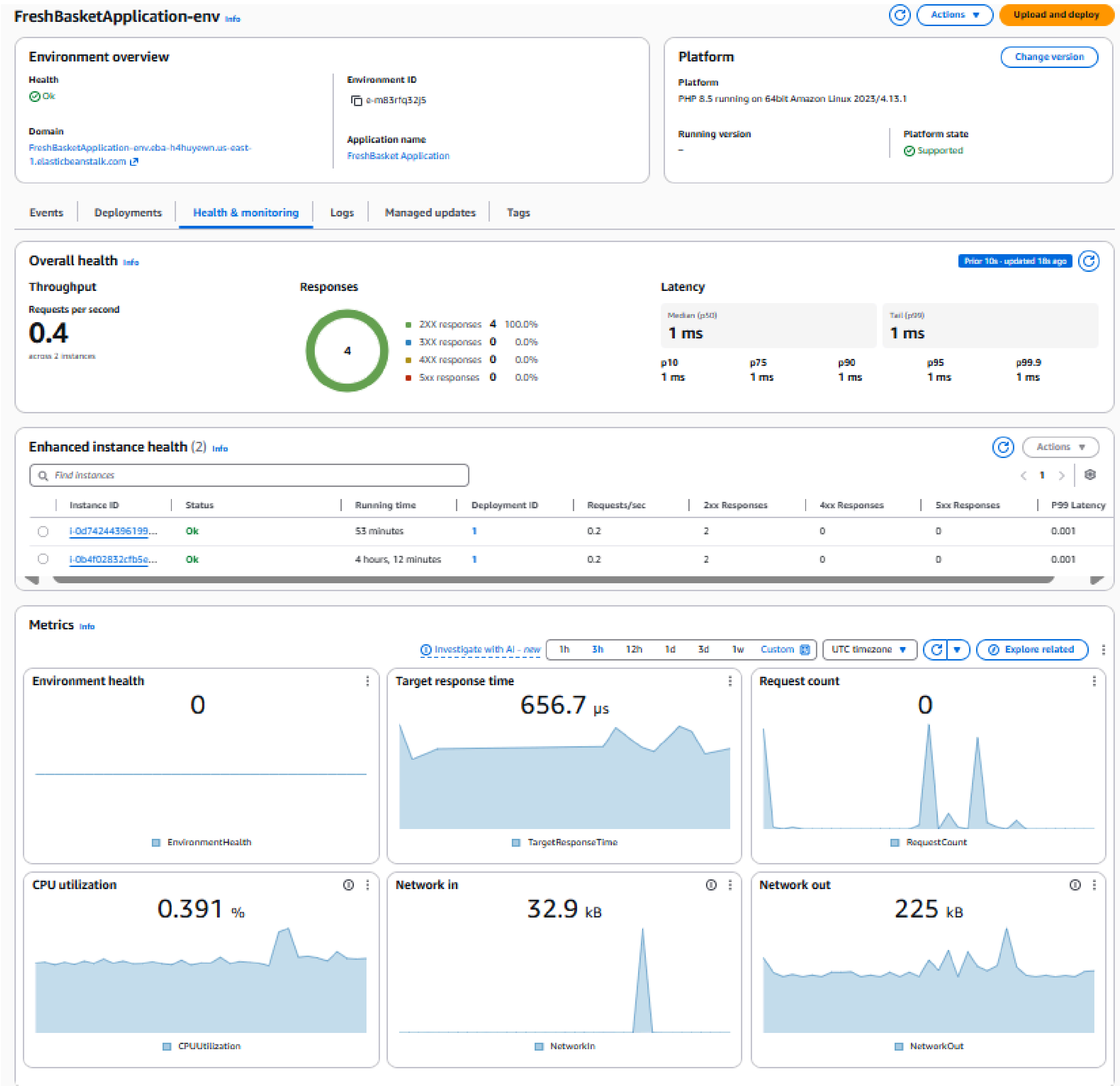


Figure 9. Configuration Panel

Configuration panel with LabRole (Service role) and LabInstanceProfile (EC2 instance profile) visible

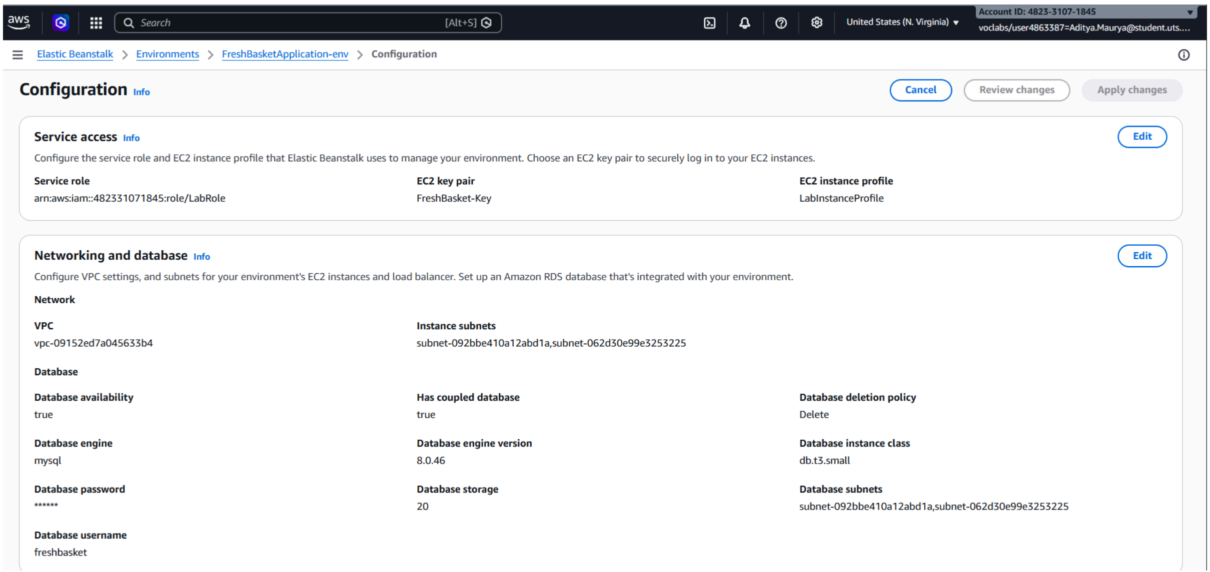



Figure 10. Auto Scaling Configuration

Beanstalk Configuration → Capacity showing min 2, max 8, CPU scale-out at 60% and scale-in at 30%.

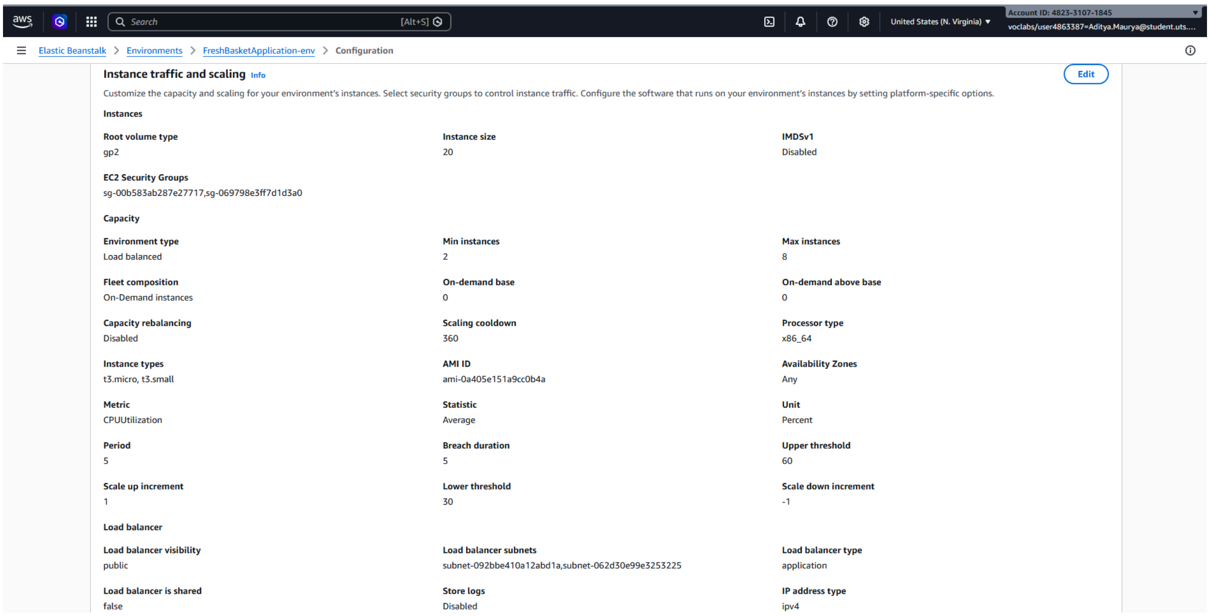
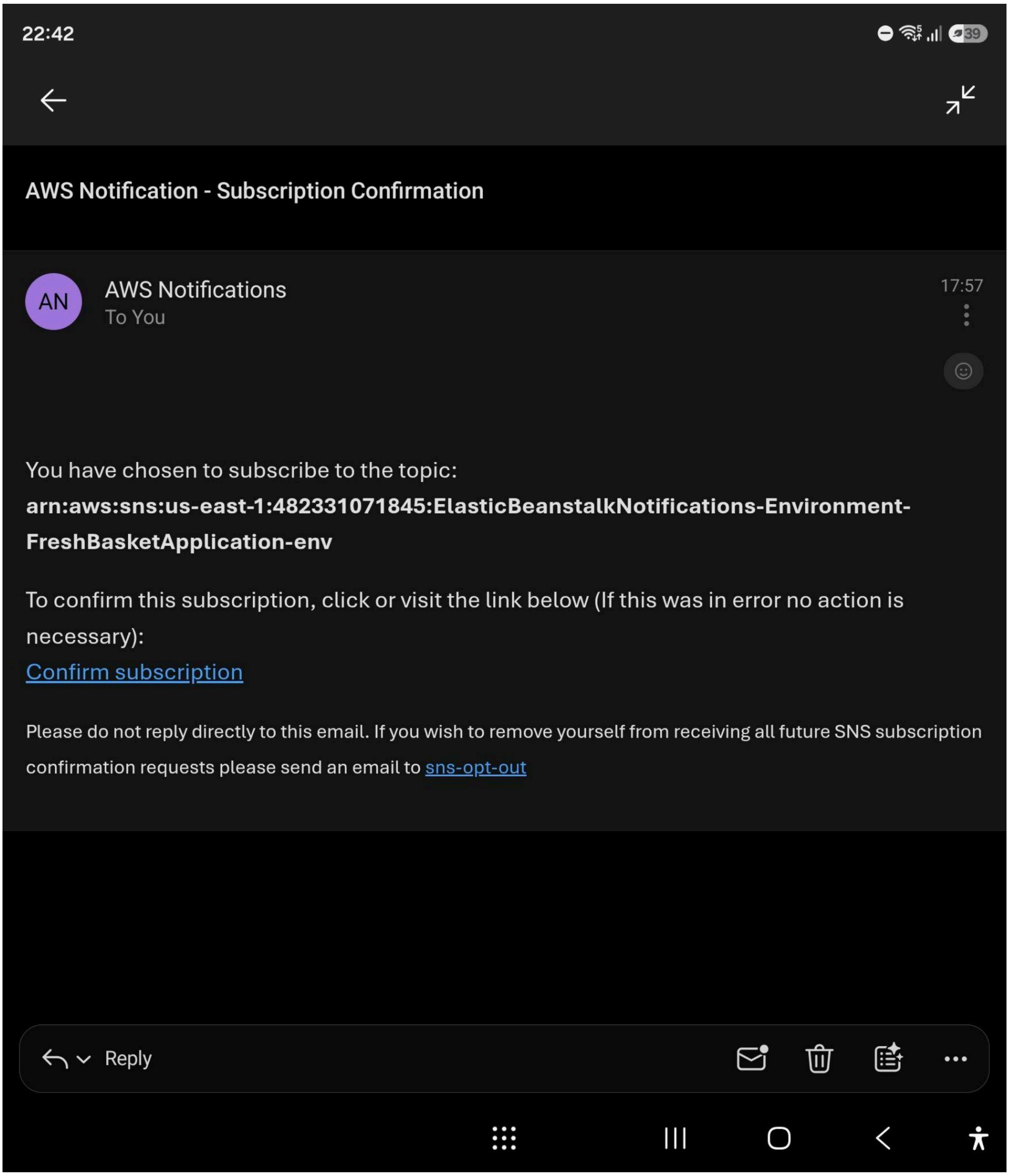


Figure 11. Email Notifications

SNS subscription confirmation email AND at least one Elastic Beanstalk event notification email, both with subject and timestamp visible.



22:41

**AWS Elastic Beanstalk Notification - Environment health has transitioned from Ok to Info. Applica...**

AN AWS Notifications
To You

22:37

Timestamp: Thu Jun 04 12:36:51 UTC 2026
Message: Environment health has transitioned from Ok to Info. Application restart in progress (running for 3 seconds).

Environment: FreshBasketApplication-env
Application: FreshBasket Application

Environment URL: http://FreshBasketApplication-env.eba-h4huyewn.us-east-1.elasticbeanstalk.com
NotificationProcessId: b22d6238-42db-4199-844d-f797902d40bf

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